

The Shrinking Edge: What Survives a Real Fill in a \$97 Billion Prediction Market

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A prediction-market price is meant to be a probability, and every gap between the two looks like money on the table. Studies of Polymarket and Kalshi keep finding such gaps: longshots that lose, a tilt toward Yes, forecasters and models that beat the price. We ask a narrower question. How much of that could a trader have collected? We assemble 1,009,373 resolved Polymarket markets (\$97 billion traded), reconstruct 23.7 million taker fills on 48,193 of them, and send each claimed edge down a ladder of controls: prices someone actually paid, estimators that average zero when prices are fair, stale information, a capital limit, fees, and a confirmation period fixed in advance. Three measurement problems account for most of what looked like mispricing. Quote midpoints in dormant books sit near 0.5 and manufacture a Yes bias of 14 points where nobody could trade. Pooling fills and taking the first fill in a price band give opposite signs for the favorite–longshot bias on the same tape. Recalibration methods that fit one period lose to the raw price in the next. Two effects survive. A textbook digital-option formula, fed only public spot data, carries as much information as the fill price on crypto threshold contracts, and the fills it endorsed returned +18.6% (+4.7 to +34.0) after fees; the edge falls to +7.2% (-0.3 to +14.3) when the formula’s spot input is fifteen minutes old, and to a level we cannot distinguish from zero in the two most recent quarters. The first print of a contract inside a high price band overshoots the outcome by about eleven points, and fading it at the next real fill returned +15.5% (+10.8 to +20.0), on fills with a median stake of \$5. A walk-forward residual model improves the Brier score of the last traded price by +1.87%. Language models with no retrieval add nothing on 282 questions opened after their training cutoffs.

1 Introduction

On the evening of January 23, 2025, a contract on Polymarket asked whether Bitcoin would be above \$103,000 at noon the next day. Bitcoin was trading at \$105,434. A textbook formula for a cash-or-nothing option, given that spot price, the strike, twenty-one hours to go, and the volatility of the previous week, valued Yes at 71 cents. A trader bought 8,680 shares of Yes at 57 cents. Twenty-one hours later Bitcoin closed above the strike and the trader collected \$3,700. Over the following hours, as the market priced Yes at 78 cents and the formula said 65, a second group of traders bought No, and the same formula would have told them to. They lost \$53,000.

Both episodes are in the public record, fill by fill, and that record is the subject of this paper. Prediction markets have grown from a curiosity into venues that traded \$97 billion on Polymarket alone, and the question of whether their prices can be trusted as probabilities has moved from seminar rooms to newspapers, central banks, and courtrooms. Economists have studied that question for twenty years and have an answer: the prices are approximately right, with systematic exceptions [Page and Clemen, 2013, Wolfers and Zitzewitz, 2004]. What has changed is that the exceptions now arrive in a stream. Recent work reports a favorite–longshot bias on Kalshi [Becker, 2026, Bürgi et al., 2026], political prices compressed toward one half [Le, 2026], a small set of accounts that drive price discovery [Gómez-Cram et al., 2026], tens of millions of dollars of arbitrage [Saguillo et al., 2025], and language models that approach the crowd forecast or price [Halawi et al., 2024, Yang et al., 2026]. Each of these is also a trading rule. If five-cent contracts win three percent of the time, selling them makes money. If a forecaster is better calibrated than the price, following the forecaster makes money.

We ask a narrower question than whether the prices are right: how much of each reported gap could a trader have collected? The question matters for an investor, who wants to know whether an edge is real before paying for it, and it matters for the literature, because an apparent gap that no trade could have exploited tells us about the measurement rather than the market. Three features of Polymarket make the question answerable. Every public price history is a quote midpoint,

which exists whether or not anyone could trade near it. A complete record of taker fills, with prices, sizes, and wallet addresses, is public for every market. And the fee schedule changed the economics of most contracts during 2026, which gives a natural confirmation period. An apparent edge can therefore be walked down a ladder: from displayed prices to prices that traded, from pooled averages to estimators that are fair under the null, from current information to stale information, from unlimited size to a capital limit, from the whole sample to a period chosen before looking. We run that ladder for each family of claims and report every rung.

The headline is that most of the mispricing goes away on the first rungs, and what remains is small, specific, and shrinking. Among quote snapshots in markets we hold fills for, 32% had no trade in the preceding day; their midpoints sit near 0.5 and resolve Yes 14 points less often than displayed. On the same fills, a share-weighted pool says that Yes prices between 0.8 and 0.9 are too low by 6.5 (3.9 to 8.6) points, while a one-bet-per-market rule says they are too high by 11.3 (10.1 to 12.4) points. Platt and isotonic recalibration, fitted walk-forward, score worse than the raw price. Two things do survive. On 9,612 crypto threshold contracts, a driftless lognormal digital-option formula with trailing realized volatility ties the fill price in Brier score and receives a weight of 0.60 (0.39 to 0.69) next to the price in a regression of the outcome on both; the fills it endorsed by at least ten cents returned +18.6% (+4.7 to +34.0) to the taker after fees, falling to +7.2% (-0.3 to +14.3) when its spot input is fifteen minutes old and to roughly zero after a seven-percent taker fee reached every crypto contract. And the first fill of a market inside a high price band overshoots the outcome by about eleven points, which a later real fill can fade for +15.5% (+10.8 to +20.0), on fills that have a median stake of \$5.

This paper makes three contributions. First, a dataset and a method. We collect every resolved Polymarket market with its rules and fee schedule, quote paths for 170 thousand of them, and 23.7 million taker fills (\$2.83 billion staked) on a stratified sample of 48,193 markets, joined to one-minute crypto spot, an implied-volatility index, and hourly equity bars, all from public endpoints. We evaluate every claim at fills, with sampling rules and estimators whose expectation is zero when fill prices are fair, and with inference that resamples events or expiry dates. Second, three measurement results: dormant midpoints manufacture a Yes bias; the sign of the favorite-longshot bias depends on whether fills are pooled or sampled once per market; and the average of per-market return ratios is biased upward under the null. Each is a mechanism rather than a correction, and each explains why published estimates on the same venue disagree. Third, an economic result about where edge lives. Polymarket lists tens of thousands of contracts that are digital options on crypto and single-stock closes, and a formula that fits nothing is as informative as the price. The edge that formula identifies decays with the age of its inputs and disappeared when fees arrived, which is the signature of a race against transaction costs rather than of risk compensation; the implied volatility backed out of Polymarket fills matches what was realized to within a few percent, so no volatility premium is being paid. Learned models, wallet records, and language models without retrieval add little or nothing.

Section 1.1 places the paper in three literatures. Section 2 describes the platform, the data, and the sample. Section 3 defines an edge at a fill and the estimators. Section 4 presents the three measurement results. Sections 5 to 7 cover calibration, option pricing, and challengers to the price. Section 8 reports what survives a capital-constrained replay, and Section 9 interprets it.

1.1 Related work

Prices as probabilities. Wolfers and Zitzewitz [2004] survey the case for reading prediction-market prices as forecasts and document their accuracy. Manski [2006] shows that with heterogeneous beliefs the equilibrium price need not equal the mean belief, and Wolfers and Zitzewitz [2006] reply that under broad conditions the price remains a useful, if sometimes biased, estimate. Page and

Clemen [2013] find prices well calibrated near expiry and biased far from it. Snowberg and Wolfers [2010] and Ottaviani and Sørensen [2015] supply the two leading explanations of the favorite-longshot bias, misperception and informational, the second derived for prediction markets. On Kalshi, Bürgi et al. [2026] report that contracts under ten cents lose most of their stake and that makers earn more than takers, and Becker [2026] reports a symmetric maker-taker transfer across 72 million trades. On Polymarket, Le [2026] estimate recalibration slopes above one in political markets, Walker [2026], in a senior thesis, reports a slope near 1.1 over 188 thousand markets, Reichenbach and Walther [2025] study accuracy and trader skill, and Cardozo and Rivero-Wildemaube [2026] show that the longshot loss changes sign when contracts are grouped by event. We add the distinction between quotes and fills, and between sampling rules, and show that it accounts for the disagreement.

Who is informed. Gómez-Cram et al. [2026] attribute price discovery to about three percent of accounts whose skill persists out of sample. Sirolly et al. [2025] estimate that wash trading reached nearly sixty percent of Polymarket volume at its December 2024 peak, and Mitts and Ofir [2026] document informed trading ahead of geopolitical, corporate, and entertainment events. We test persistence of realized wallet returns on a sample of markets and use wallet records as model features; our null on persistence is weaker evidence than the full-chain result, and we say so.

Language models as forecasters. Halawi et al. [2024] build a retrieval-and-reasoning system that approaches the crowd aggregate; Schoenegger et al. [2024] find a crowd of twelve models indistinguishable from a human crowd; Karger et al. [2025] benchmark models against expert forecasters; Yang et al. [2026] score models against live market prices and find positive returns in some settings with slower aggregation than markets near resolution. Paleka et al. [2026] and Lopez-Lira et al. [2025] document temporal leakage, including through training data. Our test uses only markets that opened after every model’s cutoff, gives no retrieval, and scores each forecast against the price at the same instant.

Event contracts and derivatives. Breeden and Litzenberger [1978] show that option prices identify a state-price density. Diercks et al. [2026] compare Kalshi macro contracts with fed funds futures, option-implied distributions, inflation swaps, and surveys. Portnaya [2026] compares three Polymarket Bitcoin contracts with option-implied values. We price 9,612 crypto threshold contracts and 2,694 single-stock contracts at every fill.

Inference for backtests. We follow Bailey and López de Prado [2014], Harvey et al. [2016], and White [2000] in discounting for the number of variants examined, and score forecasts with proper rules [Brier, 1950, Gneiting and Raftery, 2007], decomposed after Murphy [1973].

2 Setting and data

2.1 How Polymarket works

A Polymarket market is a question with a resolution rule and two tokens, Yes and No, each of which pays one dollar if its side resolves true. Tokens trade on a central limit order book in increments of a tenth of a cent. A taker whose order crosses the spread pays, since late 2025 for most categories, a fee of $\rho c(1 - c)$ dollars per share, where c is the price paid and ρ is a category rate between 0.03 and 0.07; a 50-cent crypto contract carries a fee of 1.75 cents. Makers pay nothing and in some categories receive a rebate. Outcomes are proposed and, if unchallenged, settled through an optimistic oracle, typically within hours of the event. Many questions are tied to a traded underlying: whether Bitcoin’s one-minute Binance candle at noon Eastern closes above a strike, whether NVIDIA’s official close on a given day is above a level, whether the price moves

up or down over a five-minute window. These contracts are cash-or-nothing options, and their settlement rules name the data source and the instant.

2.2 Data

Markets. From the public Gamma API we pull every closed market with at least \$1,000 of lifetime volume: 1,009,373 markets ending between 2020 and October 2026, carrying the question, the resolution rules, the scheduled end, the close time, the final payout, tags, and the taker-fee schedule. We keep markets that paid 0 or 1 and assign one of twelve categories from tags and question text (Table 1).

Quotes. The public price-history endpoint returns a midpoint series for each outcome token. We fetch it for every market outside sports, crypto up/down, and weather, and for a uniform random sample of the rest, 170 thousand markets in all, at one-minute to twelve-hour resolution depending on the market’s lifetime.

Fills. The public trade feed returns every taker fill with price, size, side, outcome token, timestamp, and wallet. We queue markets in four interleaved strata, crypto threshold contracts since 2025, finance, other non-sports categories, and a uniform random draw of everything, and stop at a fixed time. The sample holds 48,193 markets and 23.7 million fills before each market’s close, with \$2.83 billion of taker stake. Each fill is reduced to a Yes price, a direction, and a cost (Section 3).

Underlyings. One-minute Binance candles for BTC, ETH, SOL, and XRP from 2024; the Deribit DVOL implied-volatility index; daily and hourly bars for the equities named in stock-linked contracts.

Validation. We parse each price-linked question into option terms and replay its settlement rule on the underlying data. The API’s end time is date-only, or an hour off, for about three percent of crypto threshold contracts, so we read the settlement instant from the rules themselves; the replay then reproduces the official outcome for 99.97% of Binance-settled threshold contracts (99.71% with the API’s end time) and 99.9% of single-stock contracts. Contracts settled on Chainlink streams or futures replay less exactly and are analysed separately or excluded.

2.3 Sample

Table 1 summarizes the universe and the fill sample. Sports and five-to-fifteen-minute crypto contracts are three quarters of the count; politics is the second-largest category by dollars. The fill sample over-represents price-linked and long-tail categories by design, which is why every statistic below is reported within category or with one weight per market.

3 Measuring an edge at a fill

Every fill in the feed is a taker buying one token. A purchase of No at q is the same position as a sale of Yes at $1 - q$, so we write each fill as a Yes price p , a direction $d = +1$ if the taker ends long Yes and -1 otherwise, and a cost per share for the side bought,

$$c = \begin{cases} p & d = +1, \\ 1 - p & d = -1, \end{cases} \quad (1)$$

with the maker holding the other side at $1 - c$. With outcome $y \in \{0, 1\}$ the taker’s side wins $w = y$ if $d = +1$ and $1 - y$ otherwise, and the fee is $f = \rho c(1 - c)$ from the market’s own schedule. The

Table 1. The universe and the fill sample by category.

Category	Markets	Volume share (%)	Outcome 0 wins (%)	Markets with fills	Fills (M)	Taker stake (\$M)
sports	391,471	50.0	41.2	6,551	1.85	258
crypto up/down	372,397	11.6	49.9	6,112	5.85	97
crypto thresholds and other crypto	49,331	6.7	30.1	12,652	3.95	502
weather	104,545	1.1	11.9	1,781	0.45	11
finance	20,914	1.3	34.4	8,229	2.04	210
politics	18,263	18.9	23.8	3,926	4.51	1026
mentions	21,593	2.2	30.3	4,488	2.48	225
culture	11,745	1.8	20.1	2,421	0.94	161
economy	4,553	4.8	22.8	952	0.97	251
geopolitics	2,156	0.9	29.4	485	0.46	64
technology	2,239	0.3	19.1	495	0.19	18
other	1,237	0.4	36.7	101	0.01	2
all	1,000,444	100	39.7	48,193	23.72	2825

Resolved binary markets with at least \$1,000 of volume, 2020 to October 2026. Outcome 0 is Yes in Yes/No markets. The fill sample is stratified: all crypto threshold contracts after 2025 were queued first, then finance, then other non-sports categories, then a uniform draw of everything, interleaved; the download stopped at the cutoff with the sample as shown.

return on a set of fills F with sizes n_i is dollars won over dollars staked,

$$R(F) = \frac{\sum_{i \in F} n_i (w_i - c_i - f_i)}{\sum_{i \in F} n_i c_i}. \quad (2)$$

The null hypothesis throughout is that fill prices are fair: $E[y - p \mid \mathcal{F}_t] = 0$ for a fill at time t , where $\mathcal{F}_t$ is public information at t . Under the null, any rule for selecting fills that uses only $\mathcal{F}_t$ yields $R = 0$ in expectation before fees, and any forecast $\hat{p}_t$ built from $\mathcal{F}_t$ can do no better than p_t . Our tests are of two kinds. Scoring tests compare $\hat{p}$ with p at the same fills by Brier score $(\hat{p} - y)^2$ with one weight per market. Return tests apply an explicit selection rule and report R .

Two sampling rules recur. The pooled rule takes every fill in a set, weighted by shares. The first-touch rule takes one fill per market, the first to satisfy a condition, which makes it a stopping time. Both are adapted to $\mathcal{F}_t$ and both have expectation zero under the null. A third rule, the average over markets of each market's own return ratio, does not: when a market's early favorite loses, the market also collects winning fills after the other side becomes the favorite, so the ratio is diluted toward zero on losses and not on wins. We do not use it.

For model evaluation we sample each market at times fixed by its published schedule, eleven fractions of scheduled life and six fixed horizons before the scheduled end, and record the last traded price and features computable from fills up to that time. Confidence intervals resample events (groups of markets resolving on one question) or, for price-linked contracts, expiry dates, by a Poisson bootstrap; one weight is assigned per market.

4 Three ways to find an edge that is not there

4.1 Dormant midpoints

A midpoint exists whenever a book has two quotes, however far apart. A book quoted at one cent and 99 cents has a midpoint of fifty. Among quote snapshots in markets for which we hold the fill history, 32% had no trade in the preceding 24 hours. Dormant midpoints fall between 0.4 and 0.6 in 38% of cases, against 14% for active snapshots. In Yes/No markets outside sports, crypto, and weather, dormant midpoints in that zone resolve Yes 14 points less often than their level; active midpoints in the same zone are off by 7 points (Figure 1, left). Any calibration curve built on the price-history endpoint carries this Yes bias, and no trade could have collected it.

Table 2. Outcome minus Yes price at fills in long-tail Yes/No markets, two sampling rules.

Yes price band	Markets	Pooled (pts)	95% CI	First touch (pts)	95% CI
0.02-0.1	14,675	-2.2	(-3.0, -1.3)	+3.1	(+2.7, +3.6)
0.1-0.2	11,704	-2.3	(-5.1, +1.3)	+2.2	(+1.5, +2.9)
0.2-0.3	9,871	+3.4	(-2.8, +9.3)	-0.5	(-1.4, +0.4)
0.3-0.4	8,775	-2.0	(-8.1, +4.5)	-3.2	(-4.2, -2.2)
0.4-0.5	8,506	-1.9	(-9.8, +6.6)	-6.4	(-7.6, -5.3)
0.5-0.6	7,445	+0.1	(-11.7, +11.7)	-6.1	(-7.3, -5.0)
0.6-0.7	7,012	+4.4	(-1.1, +10.2)	-9.2	(-10.4, -8.1)
0.7-0.8	6,785	+10.0	(+3.9, +14.0)	-11.1	(-12.3, -9.9)
0.8-0.9	6,504	+6.5	(+3.9, +8.6)	-11.3	(-12.4, -10.1)
0.9-0.98	6,583	+0.9	(-1.8, +2.7)	-11.8	(-12.8, -10.8)

21,003 Yes/No markets outside sports, crypto, and weather. Pooled: every fill, share-weighted. First touch: the first fill of each market inside the band. Both rules use only the past, so both average zero when fill prices are fair. Intervals resample events.

The same artifact inflates model comparisons. Scored at displayed midpoints, the option-model rule of Section 6 returns +53% (+50 to +56). Scored at fills, the same rule returns +18.6% (+4.7 to +34.0).

4.2 Pooling versus stopping

Figure 1 (right) and Table 2 show the surprise $y - p$ by Yes-price band on the same fills under the two rules. Pooled, Yes prices between 0.8 and 0.9 are too low by 6.5 (3.9 to 8.6) points. One bet per market at the first fill inside the band, they are too high by 11.3 (10.1 to 12.4) points across 21,003 markets; below ten cents, the first-touch surprise is +3.1 (+2.7 to +3.6).

Both numbers are real. The pooled rule weights a market by how much it trades inside the band, and volume concentrates where a contract is on its way to resolution. The first-touch rule weights each market once, at the moment its price first enters the band, and 30% of those moments are a single order sweeping a thin book: the next five fills undo the move, and the taker paid 45 (42 to 48) points too much. Where the move is sustained over the next five fills, the level it settles at still exceeds the outcome frequency by 6.2 (2.8 to 9.4) points. The favorite-longshot bias on this venue has the sign the sampling rule gives it. One bet per market on the favorite at its first fill between 0.70 and 0.85 returns -2.65% (-3.25 to -2.03) before fees over 31,956 markets, and -3.80% (-4.18 to -3.40) between 0.85 and 0.95.

5 Calibration, and who wins

At scheduled snapshots, last traded prices have a Brier score of 0.107. We summarize calibration by the slope b in $\Pr(y = 1) = \sigma(a + b \logit p)$, where $b > 1$ means prices are not extreme enough and $b < 1$ means they are too extreme. Table 3 reports slopes below one in most categories (sports 0.85 (0.79 to 0.90), politics 0.93 (0.87 to 0.99), weather 0.75 (0.63 to 0.89)) and a slope that rises from 0.79 in the first 15% of a market's life to 0.96 in the last quarter. Early prints are noisy, and the noise looks like overconfidence. This is the opposite of the compression toward one half reported from pooled political trades, and the previous section says why: a pooled tape and a clock-sampled tape weight different moments.

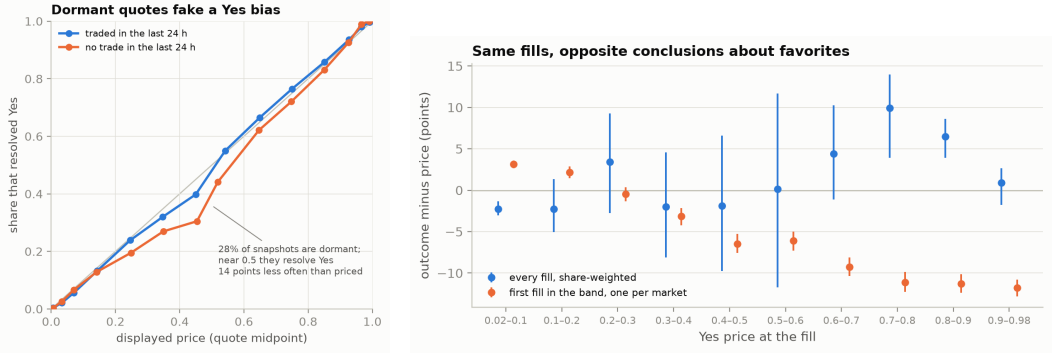


Fig. 1. Left: reliability of displayed midpoints in long-tail Yes/No markets, split by whether the market traded in the prior day. Right: outcome minus price by Yes-price band on the same fills under two sampling rules. Whiskers are 95% intervals from an event-level bootstrap.

Table 3. Calibration of the last traded price at scheduled snapshots.

Sample	Markets	Brier	Mean price	Mean outcome	Slope b	95% CI	Intercept a	95% CI
crypto thresholds and other crypto	12,273	0.070	0.326	0.316	0.93	(0.89, 0.97)	-0.17	(-0.24, -0.11)
crypto up/down	2,109	0.243	0.504	0.501	0.92	(0.73, 1.11)	-0.01	(-0.10, +0.08)
culture	2,355	0.073	0.210	0.200	0.88	(0.83, 0.95)	-0.23	(-0.37, -0.09)
economy	922	0.102	0.234	0.253	0.95	(0.85, 1.07)	+0.15	(-0.04, +0.39)
finance	7,972	0.102	0.328	0.332	0.92	(0.89, 0.96)	+0.01	(-0.06, +0.08)
geopolitics	473	0.101	0.257	0.294	1.12	(0.99, 1.28)	+0.48	(+0.20, +0.80)
mentions	4,054	0.124	0.313	0.289	0.87	(0.82, 0.92)	-0.24	(-0.33, -0.14)
politics	3,687	0.090	0.228	0.234	0.93	(0.87, 0.99)	+0.02	(-0.10, +0.15)
sports	5,232	0.170	0.415	0.414	0.85	(0.79, 0.90)	-0.04	(-0.10, +0.02)
technology	483	0.066	0.207	0.203	1.01	(0.90, 1.15)	-0.07	(-0.48, +0.26)
weather	1,776	0.080	0.131	0.126	0.75	(0.63, 0.89)	-0.38	(-0.62, -0.12)
life fraction 0-0.15	22,530	0.122	0.308	0.290	0.79	(0.77, 0.82)	-0.26	(-0.30, -0.22)
life fraction 0.15-0.45	31,028	0.112	0.311	0.297	0.90	(0.88, 0.92)	-0.17	(-0.21, -0.13)
life fraction 0.45-0.75	33,045	0.097	0.297	0.287	0.93	(0.91, 0.96)	-0.13	(-0.17, -0.10)
life fraction 0.75-1.0	36,070	0.082	0.300	0.295	0.96	(0.94, 0.99)	-0.08	(-0.12, -0.04)

Logistic recalibration $\Pr(y = 1) = \sigma(a + b \logit p)$ fitted with one weight per market. $b < 1$ means prices are too extreme. Snapshots with a trade in the prior six hours or 10% of the market's life. Intervals from 300 bootstrap draws that resample events.

Table 4 asks who wins. Takers as a group earn -0.27% (-0.61 to +0.09) after fees: we find no measurable maker-taker transfer in aggregate, in contrast with Kalshi. The exception is the five- and fifteen-minute crypto up/down contracts, where takers lose 2.34% (1.49 to 3.20) and pay 2.63% of stake in fees; there the loss is the fee.

6 Event contracts as digital options

A contract that pays one dollar if Bitcoin closes above K at time T is a cash-or-nothing call. With spot S_t , remaining time τ , and volatility σ , a driftless lognormal model values it at

$$\hat{p}_t = \Phi\left(\frac{\ln(S_t/K) - \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}\right), \quad (3)$$

where Φ is the standard normal distribution function. Range contracts are differences of two such terms, and one-touch contracts follow from the reflection principle. We set σ to realized volatility

Table 4. Taker returns on dollars staked, by category.

Category	Markets	Stake (\$M)	Gross (%)	95% CI	Net of fees (%)	95% CI	Fees (% of stake)
crypto thresholds and other crypto	12,652	502	+0.15	(-0.10, +0.39)	-0.02	(-0.27, +0.22)	0.17
crypto up/down	6,112	97	+0.28	(-0.52, +1.13)	-2.34	(-3.20, -1.49)	2.63
culture	2,421	161	-0.47	(-0.92, +0.22)	-0.54	(-0.99, +0.12)	0.07
economy	952	251	-0.53	(-1.28, +0.14)	-0.63	(-1.43, +0.05)	0.11
finance	8,229	210	+0.08	(-0.38, +0.64)	-0.16	(-0.61, +0.38)	0.25
geopolitics	485	64	-0.69	(-1.53, +0.13)	-0.69	(-1.53, +0.13)	0.00
mentions	4,488	225	-0.09	(-0.46, +0.29)	-0.21	(-0.58, +0.20)	0.11
other	101	2	-0.72	(-9.57, +6.91)	-0.82	(-9.78, +6.89)	0.10
politics	3,926	1026	-0.41	(-0.87, -0.01)	-0.44	(-0.90, -0.04)	0.03
sports	6,551	258	+1.93	(-1.02, +4.91)	+1.22	(-1.74, +4.19)	0.71
technology	495	18	-0.42	(-3.15, +1.83)	-0.66	(-3.41, +1.52)	0.24
weather	1,781	11	+1.19	(-0.18, +3.12)	+0.79	(-0.61, +2.73)	0.41
all	48,193	2825	-0.03	(-0.38, +0.33)	-0.27	(-0.61, +0.09)	0.24

Return is dollars won over dollars staked across every fill in the sample; the maker's gross return is the negative of the taker's. Intervals from 1,000 bootstrap draws that resample events.

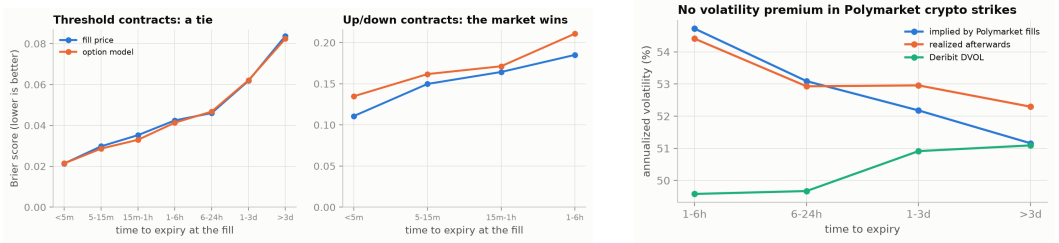


Fig. 2. Left: Brier score of the fill price and of the option model at the same fills, by time to expiry. Right: volatility implied by Polymarket fills, volatility realized from the fill to expiry, and Deribit DVOL.

over a trailing window of about four times the remaining horizon, use the last one-minute candle that closed before the fill, and fit nothing. For single stocks the clock runs in trading time: with twenty-day close-to-close volatility, each remaining overnight gap carries a quarter of a day's variance and each session the rest.

6.1 Does the formula know anything the price does not?

Table 5 scores the fill price and the formula at the same fills. Across 9,612 threshold contracts they are tied at every horizon from five minutes to a week. In a logistic regression of the outcome on the log-odds of both forecasts, the price receives weight 0.52 and the formula 0.60 (0.39 to 0.69). Had the price already contained the formula, the second weight would be zero. On short-dated up/down contracts, by contrast, the weight on the formula is 0.07 (-0.03 to 0.20): that market already knows everything the formula does.

Inverting Equation 3 at each fill gives the volatility Polymarket is pricing. Its ratio to the volatility realized between the fill and expiry lies between 0.97 and 0.99 at every horizon (Figure 2, right). Listed options usually price volatility above what is realized; these contracts do not, which is one piece of evidence that what follows is not risk compensation.

6.2 Fills the formula endorses

Call a fill endorsed when the formula values the side the taker bought at least ten cents above the price paid plus the fee, and opposed in the mirror case. Table 6 reports taker returns on both

Table 5. Fill price against the option model, scored at the same fills.

Time to expiry	Markets	Fills	Brier, fill	Brier, model	Difference	95% CI
<5m	2,533	19,654	0.0214	0.0214	+0.0000	(-0.0023, +0.0022)
5-15m	3,792	42,309	0.0298	0.0287	-0.0011	(-0.0048, +0.0014)
15m-1h	6,679	175,326	0.0352	0.0330	-0.0023	(-0.0060, +0.0001)
1-6h	9,154	751,022	0.0424	0.0413	-0.0011	(-0.0036, +0.0005)
6-24h	9,296	1,478,997	0.0460	0.0468	+0.0007	(+0.0002, +0.0013)
1-3d	9,612	864,966	0.0618	0.0621	+0.0003	(-0.0003, +0.0009)
>3d	8,503	441,346	0.0838	0.0824	-0.0014	(-0.0023, -0.0006)
<i>Up/down contracts</i>						
<5m	5,751	2,750,115	0.1107	0.1348	+0.0241	(+0.0189, +0.0278)
5-15m	2,697	1,207,764	0.1498	0.1617	+0.0120	(+0.0092, +0.0152)
15m-1h	773	348,903	0.1643	0.1713	+0.0070	(+0.0017, +0.0138)
1-6h	121	14,381	0.1850	0.2108	+0.0258	(-0.0033, +0.0601)

Threshold contracts in the top block. One weight per market; intervals from 600 bootstrap draws that resample expiry dates. Negative differences favor the model.

Table 6. Taker return after fees on crypto threshold fills the option model endorses or opposes.

Spot age	Bar	Markets	Endorsed stake (\$M)	Return (%)	95% CI	Opposed stake (\$M)	Return (%)	95% CI
1 min	0.05	4,699	26.3	+5.8	(-2.4, +13.5)	36.2	-3.4	(-9.4, +3.0)
1 min	0.10	2,415	7.0	+18.6	(+4.7, +34.0)	11.5	-10.2	(-18.2, -1.9)
5 min	0.05	4,857	30.5	+4.4	(-2.0, +11.0)	48.7	-2.7	(-7.2, +2.3)
5 min	0.10	2,987	11.9	+8.7	(-0.6, +17.6)	21.8	-4.8	(-10.1, +0.8)
15 min	0.05	5,117	35.7	+4.2	(-1.1, +9.8)	56.5	-2.6	(-6.6, +1.5)
15 min	0.10	3,437	16.8	+7.2	(-0.3, +14.3)	29.4	-3.9	(-8.4, +0.5)
60 min	0.05	5,544	42.9	+0.9	(-4.3, +6.0)	69.0	+0.1	(-3.5, +3.8)
60 min	0.10	4,092	23.9	+0.8	(-6.9, +7.6)	42.7	+0.5	(-3.8, +4.9)

A fill is endorsed when the model values the side bought at least the bar above its price plus the fee, and opposed in the mirror case. Spot age is how old the model's last candle is. Intervals resample expiry dates.

sets by the age of the formula's spot input. With a one-minute-old candle, endorsed fills in 2,415 markets (\$7.0 million staked) returned +18.6% (+4.7 to +34.0) after fees and opposed fills -10.2% (-18.2 to -1.9). With spot data that is five, fifteen, and sixty minutes old, endorsed fills return +8.7% (-0.6 to +17.6), +7.2% (-0.3 to +14.3), and +0.8% (-6.9 to +7.6). Most of the edge is a race to react to spot. At five and fifteen minutes the point estimates stay positive with intervals that reach zero; at an hour nothing is left.

The same construction on 2,694 contracts on the closes of large U.S. equities, with hourly bars, gives the formula a weight of 0.37 (0.27 to 0.50). Takers who trade against it by ten cents lose 23.2% (13.2 to 32.1), and still lose 18.8% (10.1 to 26.5) when the formula's bar is four hours older. Figure 3 shows both decays, by data age and by calendar quarter.

7 Can anything else beat the price?

Protocol. Every challenger is evaluated walk-forward on the snapshot panel. Test blocks are two calendar months, and a model tested in a block trains only on snapshots from markets that had resolved before the block began, so every training label was public when the forecast is made. The

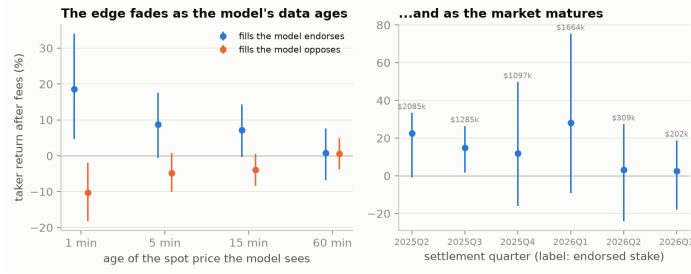


Fig. 3. Taker return after fees on crypto threshold fills that the option model endorses or opposes by at least ten cents. Left: by age of the model's spot input. Right: endorsed fills by settlement quarter, with the stake they represent.

Table 7. Brier skill of each challenger relative to the last traded price, walk-forward.

Challenger	All months	95% CI	May to Sep 2026	95% CI	Price-linked	95% CI
Quote midpoint	+0.20	(-0.35, +0.74)	+1.71	(+0.90, +2.54)	-3.11	(-4.48, -1.66)
Ten-fill average price	-0.95	(-1.29, -0.58)	-1.48	(-2.00, -1.00)	-1.77	(-2.58, -0.98)
Platt recalibration	-0.11	(-0.23, +0.02)	-0.41	(-0.59, -0.21)	+0.39	(+0.19, +0.57)
Platt by category	-0.03	(-0.21, +0.16)	-0.32	(-0.59, -0.07)	+0.39	(-0.02, +0.76)
Isotonic recalibration	-0.13	(-0.23, -0.02)	-0.29	(-0.44, -0.13)	+0.05	(-0.14, +0.24)
Residual trees: path, flow, wallets	+0.84	(+0.56, +1.09)	+1.08	(+0.75, +1.41)	+0.66	(+0.11, +1.20)
Residual trees plus option value	+1.87	(+1.51, +2.21)	+2.19	(+1.69, +2.71)	+4.97	(+3.90, +6.08)
Price and option value blend	+1.96	(+1.59, +2.32)	+1.47	(+0.92, +2.04)	+8.29	(+6.93, +9.75)
Option value alone					+7.64	(+6.21, +9.33)

Skill is the percentage reduction in Brier score relative to the price; positive beats the price. All months: 36,398 test markets, price Brier 0.1124. Last five months: 16,325 markets. Price-linked: 13,357 crypto and single-stock threshold markets. Two-month test blocks; a model tested in a block trains only on markets that resolved before it. Intervals resample events.

baseline is the last traded price. Table 7 reports Brier skill relative to it over 36,398 test markets, separately for the last five months (May to September 2026), and on price-linked contracts. No feature or setting was chosen on the last five months. They are not a pristine holdout, because we saw scores for the same configurations on those months on an earlier tape half the size; afterwards we changed only the block length and the number of boosting rounds, to save compute.

Recalibration. Platt scaling of the price, isotonic regression, and category-specific slopes all score at or below the raw price out of time. The miscalibration they fit in one period is absent in the next.

Learned residuals. A gradient-boosted model of the residual $y - p$ returns the price when its features carry no signal. With price-path, order-flow, and wallet-record features it improves on the price by +0.84% overall and +1.08% in the last five months. Adding the option value gives +1.87% and +2.19%. A two-parameter logistic blend of price and option value, which leaves every other contract at the price, gives +1.96% and +1.47%. On price-linked contracts the blend improves on the last trade by +8.3%, and on the quote midpoint at the same instant by +11.5%; the bare formula beats the displayed quote by +10.9%. Read as a forecast, the formula is better than the screen. By category the full model improves on the price in crypto thresholds, finance, culture, geopolitics, and mentions markets, matches it in sports, politics, economics, technology, and weather, and is worse on crypto up/down contracts.

Table 8. Language-model forecasts on 282 questions opened after the models’ training cutoffs.

Model	Sees	Brier	vs market	95% CI	Corr. with price	Weight given price	95% CI
Claude Haiku 4.5	question only	0.246	+0.0933	(+0.0618, +0.1255)	0.26	+0.24	(-0.10, +0.61)
Claude Haiku 4.5	question and price	0.152	-0.0003	(-0.0013, +0.0008)	1.00	+0.59	(-1.71, +1.76)
Claude Sonnet	question only	0.235	+0.0822	(+0.0522, +0.1140)	0.55	+0.10	(-0.20, +0.41)
Claude Sonnet	question and price	0.155	+0.0021	(-0.0008, +0.0053)	1.00	+0.54	(-0.68, +1.69)
Claude Fable 5.1	question only	0.236	+0.0831	(+0.0538, +0.1131)	0.63	+0.08	(-0.24, +0.41)
Claude Fable 5.1	question and price	0.154	+0.0014	(-0.0006, +0.0035)	1.00	+0.92	(-0.68, +2.09)

Market Brier 0.153; a constant forecast at the base rate scores 0.243. The weight is the coefficient on the model’s log-odds in a logistic regression of the outcome on the log-odds of the price and of the model. Intervals resample questions, one per event.

Wallet records. For each fill we compute the taker wallet’s realized return on markets that had resolved before the fill. Fills by wallets with no resolved history return -1.14% (-2.07 to -0.05) after fees and fills by seasoned wallets +0.55% (+0.17 to +0.98). Within seasoned wallets, sorting by past return does not sort future return. Our tape covers a sample of markets, so each wallet’s record is partial; this null is weaker evidence than the full-chain result of Gómez-Cram et al. [2026].

Language models. We select 282 Yes/No markets that opened after 10 July 2026, one per event, across ten categories, and fix the forecast date at 30% of each market’s scheduled life. Three models (Claude Fable 5.1, Claude Sonnet, Claude Haiku 4.5) receive the question, the resolution rules, and the date, with no tools and no retrieval; Appendix A gives the prompts and the protocol. Table 8 reports the results. Blind, the models score 0.236, 0.235, and 0.246, against 0.153 for the market and 0.243 for a constant forecast at the base rate. Shown the market price, they return it almost unchanged, and their weight next to the price in a regression that includes both is indistinguishable from zero. Without news access a language model knows the base rate and nothing the market does not.

8 What survives

8.1 A capital-constrained replay

Table 9 replays the option-model rule on real fills with a \$200 cap per fill and profit booked at expiry. The ten-cent bar was chosen on contracts that expired before April 2026; later contracts form a confirmation period. With fifteen-minute-old spot, the crypto rule returns +8.1% on stake over the whole sample with at most \$379 thousand deployed, an annualized Sharpe ratio of 1.33 on daily returns ($t = 1.76$), and a deflated Sharpe probability of 0.38 after discounting for the 24 variants we examined. The pooled fill-level interval excludes zero; the daily series does not. In the confirmation period the rule returns +1.5% ($t = 0.33$). Two things changed in that period: a seven-percent taker fee rate reached every crypto contract in the second quarter of 2026, and the stake clearing a ten-cent bar fell by a factor of five or more. In 2026Q3, endorsed fills return +2.6% (-17.9 to +18.8). The single-stock rule is smaller and cleaner: +15.5% on stake with at most \$43 thousand deployed, a Sharpe ratio of 2.53, a deflated Sharpe probability of 0.72, +20.1% before April 2026 and +7.9% after.

8.2 First touch

The overshoot of Section 4 can only be traded at prices that exist later. After the first fill of a market inside a Yes-price band, we buy No at the next fill in which a taker bought No at least sixty seconds later, provided the Yes price is still within five cents of the band. The median wait is 73 minutes. Net of fees the fade returns +15.5% (+10.8 to +20.0) on stake for the band 0.6 to 0.8 (5,282 markets)

Table 9. Capital-constrained replay of the option-model rule, \$200 cap per fill.

Rule	Period	Markets	Stake (\$k)	Profit (\$k)	Return (%)	Peak capital (\$k)	Sharpe	<i>t</i>	DSR
Crypto, 1-min spot	all	2,415	2971	+663	+22.3	371	1.30	1.71	0.30
Crypto, 1-min spot	before Apr 2026	1,617	2673	+660	+24.7	371	1.54	1.70	0.29
Crypto, 1-min spot	Apr to Sep 2026	798	297	+3	+1.0	13	0.15	0.11	0.03
Crypto, 15-min spot	all	3,437	7733	+624	+8.1	379	1.33	1.76	0.38
Crypto, 15-min spot	before Apr 2026	2,332	6561	+607	+9.2	379	1.55	1.73	0.36
Crypto, 15-min spot	Apr to Sep 2026	1,105	1172	+18	+1.5	50	0.45	0.33	0.05
Stocks, current bar	all	1,257	278	+48	+17.3	42	2.74	2.62	0.83
Stocks, current bar	before Apr 2026	716	176	+38	+21.6	42	4.67	2.97	0.95
Stocks, current bar	Apr to Sep 2026	541	102	+10	+9.8	14	1.07	0.77	0.07
Stocks, hour-old bar	all	1,310	317	+49	+15.5	43	2.53	2.41	0.72
Stocks, hour-old bar	before Apr 2026	744	199	+40	+20.1	43	4.12	2.62	0.82
Stocks, hour-old bar	Apr to Sep 2026	566	118	+9	+7.9	17	0.96	0.69	0.07

Profit is booked on the expiry day; daily return is profit over the largest stake outstanding on any day; Sharpe is annualized. *t* is the daily-series *t*-statistic. DSR is the deflated Sharpe probability against the best of 24 skill-free variants. The ten-cent bar was fixed on contracts expiring before April 2026.

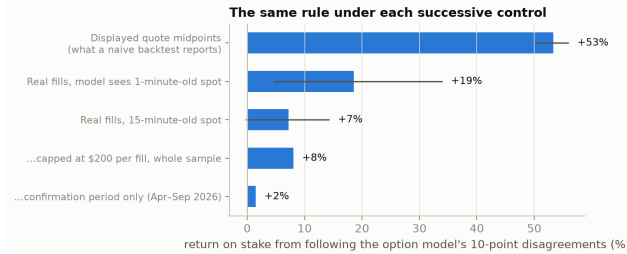


Fig. 4. Return of one rule, “follow the option model when it disagrees with the price by ten cents,” under successive controls.

and +23.4% (+15.4 to +31.9) for 0.8 to 0.98 (4,598 markets); from April 2026 onward, +16.7% (+10.7 to +22.8) and +31.8% (+20.7 to +43.7). Dropping the trigger and taking each market’s first No-buying fill in the same range returns +17.5% (+14.1 to +20.9) and +38.3% (+31.7 to +45.5), so the result does not depend on reacting to the first print. Its limit is size: the fills copied have a median stake of \$5 and total \$176 thousand across both bands. This is a pattern in thin books early in a market’s life, and it has almost no capacity.

9 Discussion

What kind of mispricing is it? Three explanations are usually on offer for a gap between a price and an outcome frequency: compensation for risk, information not yet in the price, and limits to arbitrage. The option-model results discriminate between them. Risk compensation would show up as a volatility premium, and the implied volatility in Polymarket fills matches realized volatility to within a few percent. Information not yet in the price would not decay with the age of the formula’s inputs, and it does, from +18.6% (+4.7 to +34.0) at one minute to nothing at an hour. What remains is a race: traders who could price the contract from spot faster than the book updated collected a few points per fill, and a fee of under two cents a share was enough to close most of it. The first-touch overshoot points the same way. It is a transfer from takers who sweep thin books to the makers resting there, it is largest at the first print, and the fills that carry it are tiny. Neither effect looks like a belief that is wrong; both look like liquidity that is expensive at particular moments.

Magnitudes. The endorsed-fill stake in our sample is \$7.0 million over roughly eighteen months, and the sample holds about a third of the threshold contracts listed in that period, so the platform-wide stake that cleared a ten-cent bar was on the order of twenty million dollars, and the profit on it, at +18.6% (+4.7 to +34.0), a few million dollars in total, or a few hundred thousand dollars a month before fees arrived. The capped replay never deployed more than \$379 thousand. These are the amounts available to the fastest participant, not to a fund.

Limitations. Fills are a stratified sample, so wallet histories are partial and the wallet results are weak. The trade feed reports the taker side only, and we observe fills rather than order books, so the capacity of each rule is bounded above by the sizes that traded and the fill a rule “takes” was in fact taken by someone else. Spot inputs for stock contracts are hourly and blind to extended-hours trading. Contracts settled on Chainlink streams are modeled with Binance prices. The confirmation period is six months, and the daily-series inference for the replay is underpowered at that length. The trailing-volatility rule in Equation 3 is crude and lags after a spike; the contract that opens this paper is a case where it cost the rule \$53,000, and nothing about it was tuned. For the first-touch result to be a measurement artifact, the trade feed would have to misassign the Yes and No tokens on roughly one fill in ten in a way correlated with outcomes, which the settlement replay rules out for price-linked contracts and for which we have no mechanism elsewhere. Literature values quoted in Section 1.1 are as reported by their authors.

Conclusion. Most of the mispricing measured on the largest prediction market is produced by how it is measured: dormant quotes, pooled tapes, and ratios that cannot average to zero. What remains is concentrated where a public model of the underlying exists and where thin books let one order move the price, and both remainders are shrinking. A reader who wants to know whether a prediction-market price is a probability can take it as one, with the proviso that the number on the screen is a probability only where someone has recently been willing to trade at it.

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A Language-model protocol

Questions. All Yes/No markets that opened on or after 10 July 2026 and closed by 30 September 2026, with at least \$20,000 of volume and a scheduled life of at least three days, one market per event, stratified over ten categories; 481 were selected and 282 had at least five fills before the

forecast instant and are scored. The forecast instant is 30% of the way through each market’s scheduled life; the market price at that instant is the size-weighted mean of the last ten fills within twelve hours.

Models and sampling. Claude Fable 5.1 (claude-fable-5-1), Claude Sonnet (the current Sonnet model, version not pinned by the interface), and Claude Haiku 4.5 (claude-haiku-4-5-20251001), each called once per batch of about 35 questions through an agent interface with default sampling settings and no tools beyond reading the question file and writing the answer file. Every run was checked to have made exactly those file operations. One sample per question; no prompt-wording robustness was run.

Prompt, condition A.. “You are a forecaster in a controlled experiment. Read the JSON file [path]. It holds binary prediction-market questions. For each item, ‘today’ is the date you are forecasting on. Estimate the probability that the question resolves YES under its resolution rules. Strict rules: use NO tools except one Read of that file and one Write of your answers. Do not search the web, fetch pages, run shell commands, or open any other file. Use only your own knowledge and reasoning (base rates, time remaining until scheduled_end, the exact resolution rules). Write a JSON list to [path] with one object per question: {“qid”: ..., “p”: probability of YES between 0.01 and 0.99}. Include every qid exactly once.” Each item carried the question, the resolution rules, the opening date, the forecast date, and the scheduled end.

Prompt, condition B.. Identical, with each item also carrying current_market_price_of_yes and the sentence “You may rely on the market price as much or as little as you judge best; your goal is the most accurate probability.”

B Fee schedule and categories

Taker fees per share are $\rho c(1 - c)$ with ρ read from each market’s schedule: 0.07 for crypto, 0.05 for sports, economics, culture, and weather, 0.04 for politics, finance, mentions, and technology, 0.03 for an earlier sports schedule, and zero for geopolitics and for markets listed before fees were introduced. Categories are assigned from Polymarket’s tags in a fixed priority order (weather, mentions, crypto, sports, economics, finance, politics, geopolitics, technology, culture), with “crypto up/down” for markets whose outcomes are Up and Down.

C Disclosure

The data pipeline, experiments, figures, and this draft were produced with the assistance of an AI coding agent operated by the author, who directed the work and is responsible for its content. The language-model forecasts in Section 7 were generated by that agent’s models under the protocol in Appendix A.